Check for updates

Advanced Theory and Simulations

ADVANCED THEORY AND SIMULATIONS
www.advtheorysimul.com

RESEARCH ARTICLE

Machine Learning Meets Molecular Dynamics: Accurate Prediction of Polymer Solubility Parameters

Alan George | Marek Sierka

Otto Schott Institute of Materials Research, Friedrich Schiller University, Jena, Germany

Correspondence: Marek Sierka (marek.sierka@uni-jena.de)

Received: 26 September 2025 | **Revised:** 4 February 2026 | **Accepted:** 18 March 2026

Keywords: feature engineering | Hildebrand solubility parameter | machine learning | MD simulations | solubility prediction

ABSTRACT

The Hildebrand solubility parameter is a cornerstone descriptor for predicting thermodynamic compatibility in polymer-solvent systems, yet its determination for polymers remains experimentally challenging, and traditional group contribution estimates lack accuracy and transferability. Here, we present an integrated framework that combines atomistic molecular dynamics (MD) simulations with machine learning (ML) to predict solubility parameter values across diverse molecular and polymeric systems. Solubility parameter values derived from MD simulations provide training targets for ML models built on molecular, quantum-chemical, and chain-specific descriptors. By systematically applying feature-selection strategies, Pearson correlation filtering, principal component analysis, and recursive feature elimination, we construct optimized descriptor subsets tailored to each learning algorithm. Benchmarking across regression methods reveals that Gaussian process regression with recursive feature elimination achieves the best performance, with $R^2 = 0.94$, MAE = 0.54, and RMSE = 0.96 $MPa^{1/2}$ on the test set, consistently outperforming both empirical group-contribution methods and conventional regressors. Our results highlight the critical role of descriptor curation and nonlinear models in capturing the complexity of polymer thermodynamics. This hybrid MD-ML strategy establishes a generalizable and computationally efficient pathway for predicting solubility parameters, enabling rapid screening of solubility and miscibility for both known and novel molecules and polymers.

1 | Introduction

The Hildebrand solubility parameter (SP) δ is a widely used, conceptually straightforward measure for evaluating the thermodynamic compatibility between two substances. It is particularly useful for predicting polymer–solvent solubility and miscibility of small molecules in the context of nonpolar or weakly polar systems [1]. The parameter is rooted in the concept of cohesive energy density (CED), which quantifies the energy required to separate molecules from each other in the bulk phase. CED can be derived from the molar enthalpy of vaporization (ΔH_{vap}), corrected for the energy of thermal motion, and reflects the strength of intermolecular forces. The Hildebrand parameter is defined as [2]

$$\delta = \sqrt{CED} = \sqrt{\frac{\Delta H_{vap} - RT}{V_{mol}}} \tag{1}$$

where ΔH_{vap} is the enthalpy of vaporization, R is the gas constant, T is the temperature, and V_{mol} is the molar volume. Comparing the δ values of two substances provides a quick heuristic to assess their mutual solubility: systems with similar values are generally miscible ("like dissolves like").

Although SP is most commonly used for evaluating compatibility between nonpolar polymers and solvents [2], its utility extends to

© 2026 Wiley-VCH GmbH

more complex systems such as polymer blends, where predicting miscibility is more challenging. In such blends, the entropy of mixing is typically low due to the large molecular sizes, and even modest unfavorable enthalpic interactions can lead to phase separation. The Hildebrand parameter provides a first-order approximation, and despite its limitations (e.g., neglecting polar and hydrogen bonding interactions), it serves as a foundation for estimating more advanced thermodynamic quantities. In current polymer research and applications, δ continues to be widely used across the full polarity spectrum of polymers. Venkatram et al. compiled experimental solvent/non-solvent data for 75 polymers, including strongly polar systems such as poly(vinyl alcohol), and showed that δ still provides useful predictive power—particularly for identifying non-solvents—across the full polarity range [3]. In addition, because Hildebrand parameters are available for a much larger experimental polymer data set than, for example, Hansen parameters, the Polymer Genome platform [4] uses δ values to train ML models and to screen solvent/non-solvent pairs for thousands of polymers. In line with this established usage in contemporary polymer informatics, we include both polar and non-polar polymers in our dataset while acknowledging the inherent constraints of the single-parameter δ framework.

Notably, the Flory-Huggins interaction parameter χ [5, 6] can be approximated using solubility parameters through the following relationship [7, 8]:

$$\chi = \frac{V_{\mathrm{m}}}{RT}\left(\delta_1 - \delta_2\right)^2 \tag{2}$$

where V_{m} is a reference volume (commonly the molar volume of the solvent or a mean segmental volume), R is the gas constant, T is the temperature, and δ_1, δ_2 are the solubility parameters of the two components.

This expression can then be inserted into the Flory-Huggins theory to estimate the Gibbs free energy of mixing; for example, in the case of polymer solutions, the equation becomes [9–11]:

$$\Delta G_{\mathrm{mix}} = RT\left[\frac{\phi_1}{N_1}\, ln\phi_1 + (1-\phi_1)\, ln(1-\phi_1) + \phi_1(1-\phi_1)\chi\right] \tag{3}$$

where R is the gas constant, T is the temperature, ϕ_1 the volume fraction of the polymer, N_1 the degree of polymerization of the polymer, $1-\phi_1$ the volume fraction of the second component. These relationships demonstrate that while solubility parameters do not capture all interaction specifics, they are integral to many practical models of polymer solution and blend thermodynamics [12, 13].

For small molecules, ΔH_{vap} can be determined experimentally. However, such measurements are infeasible for polymers due to their high molecular weights and thermal instability. As a result, indirect experimental techniques, including hot-stage microscopy, differential scanning calorimetry (DSC), and UV spectroscopy, are employed to estimate the solubility parameters of polymers [14, 15]. These methods, however, are typically limited to a narrow range of polymer classes and often suffer from uncertainties in interpretation and reproducibility.

To overcome experimental limitations, empirical estimation methods and computational simulations are increasingly employed. Among empirical approaches, the group contribution (GC) method is widely used. It estimates SP values by summing cohesive energy contributions from predefined structural fragments [16]. While computationally inexpensive, GC methods rely heavily on empirical assumptions and are limited by the availability and accuracy of fragment-specific data. Despite ongoing efforts to refine group contribution schemes, a universally accurate and transferable predictive model for diverse polymer classes remains elusive [17].

As an alternative, atomistic molecular dynamics (MD) simulations using classical force fields have gained traction as a means of evaluating solubility parameters [18–20]. For example, atomistic MD simulations were used to compute solubility parameters for indomethacin and excipients and to predict drug–carrier miscibility [18]. A cohesive-energy-density MD protocol was applied to obtain Hildebrand and Hansen parameters for organic and polymer systems [19], while Hansen parameters derived from MD were used to predict siloxane surfactant adsorption on silica and to guide solvent selection [20]. These simulations offer molecular-level insights into polymer microstructure, conformational behavior, and thermodynamic properties [21]. With the advent of accurate ab initio-derived force fields [22], MD has also been used to investigate the compatibility of polymers with small molecules and other polymers [18]. However, such simulations are computationally expensive, especially when applied to large or long-chain polymer systems [23].

More recently, machine learning (ML) techniques have emerged as promising tools for predicting solubility parameters and other molecular properties [24–26]. ML models offer the advantages of being fast, scalable, and often achieving accurate predictions, provided that sufficiently high-quality training data is available [25, 27]. In particular, ML approaches have also been successfully applied to predict χ for polymer–solvent mixtures, showing their ability to capture thermodynamic interactions [28–30]. For example, Hu et al. [31] developed a Hildebrand-parameter model using ensemble learning on the experimental DIPPR 801 database [32], employing RF and XGBoost to construct quantitative structure–property relationship models.

Flory–Huggins χ, Hildebrand $\Delta\delta$, and the relative energy difference determined from Hansen solubility parameters between polymers and solvents were investigated using interaction data from 81 polymers and 1221 solvents [29]. They trained a LightGBM model (a gradient-boosted tree-based algorithm) on SMILES-encoded trimers and solvent descriptors, identifying key nonlinear features such as LogP, molar volume, and dipole moment.

Ethier et al. [33] reviewed advances in phase-diagram prediction, thermodynamic modeling, and solubility prediction. They highlighted a multitask neural-network model [30], which outperforms COSMO-RS and Hansen-parameter methods in predicting χ values and solubility labels. They also discussed hybrid strategies such as the Gaussian-process Hansen solubility parameter (gpHSP) model [26], which integrates COSMOtherm-derived molecular information with quantum-chemical descriptors to enhance solubility-parameter prediction.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Beyond these specific studies, ML methods are also increasingly used throughout the polymer development pipeline, aiding characterization, property simulation, optimization, and synthesis [34]. Neural-network models trained on large polymer–solvent compatibility datasets have demonstrated a strong ability to leverage polymer structural features to accurately identify suitable solvents [35, 36].

Despite these advances, a consistent limitation across existing ML studies is the lack of high-quality, internally consistent experimental data [18], especially for polymers. Reported solubility parameters vary widely because polymer systems are inherently stochastic and hierarchical, and their morphology depends strongly on preparation and processing conditions. As highlighted by Martin and Audus [37], the ML pipeline is constrained at the data-collection stage by both limited data quantity and inconsistent data quality, often necessitating in-house measurements or extensive literature mining. Data-fusion approaches [38] can expand the dataset size, but they frequently introduce inconsistencies due to differing experimental protocols, further complicating model development [39]. Although databases such as PoLyInfo [40], PI1M [41], Khazana [42], PubChem [43], and OpenPoly [44] provide valuable resources, they differ in data-collection methods and in the level of polymer-specific information they contain, resulting in heterogeneous and sometimes incomplete inputs for ML modeling.

To address this gap, we propose combining atomistic MD simulations with data-driven ML models to construct an integrated predictive framework for solubility parameters. Polymers are particularly amenable to ML-based modeling due to their repetitive chain structure, which introduces both regularity and variability—an ideal feature space for supervised learning. In previous work, our group trained ML models to predict ΔH_{vap} for polymers by using small-molecule analogs structurally related to polymer repeat units [25]. These models, based on molecular descriptors alone, achieved reasonable accuracy but could not fully account for the influence of chain architecture. Recent MD–ML studies have shown that MD-derived structural and energetic features can improve solubility prediction for small molecules. In one such study, MD simulations were performed for 211 drug molecules, and MD-derived properties were used together with ensemble ML models to predict aqueous solubility [45].

In this study, we use MD-derived solubility parameters as target values for training ML models. The dataset comprises 79 polymers spanning a wide range of structural classes, including synthetic, biodegradable, thermoplastic, elastomeric, copolymeric, and specialty polymers, with δ values ranging from 11.8 $MPa^{1/2}$ to 26.6 $MPa^{1/2}$, as well as 141 small molecules, including hydrocarbons, acids, alcohols, ketones, amines, aldehydes, nitriles, chlorinated organics, and aromatic compounds, with δ values ranging from 13.7 $MPa^{1/2}$ to 38.2 $MPa^{1/2}$. Each entry is annotated with molecular, quantum-chemical, and chain-related descriptors. To optimize input features, we employed feature selection techniques such as the Pearson correlation coefficient (PCC) [46], principal component analysis (PCA) [47], and recursive feature elimination (RFE) [48].

We benchmarked a suite of regression models, including support vector regression (SVR) [49], random forest (RF) [48], least absolute shrinkage and selection operator (LASSO) [50], Gaussian process regression (GPR) [51], and neural network (NN) [52], to predict solubility parameters of both polymers and small molecules. These five models were selected because they represent complementary learning paradigms widely used in polymer-property prediction—spanning kernel-based, linear-regularized, ensemble, Bayesian, and neural approaches.

2 | Computational Details

2.1 | Software

All MD simulations were carried out in Materials Studio (Version 2022) [53] using the COMPASS III force field [54]. Density functional theory (DFT) calculations were performed employing the Turbomole program package [55]. ML models were implemented in Python 3.8 using scikit-learn (v1.7.0) [56], numPy (v1.26.4) [57], pandas (v2.3.3) [58], matplotlib (v3.8.0) [59], and sciPy (v1.15.3) [60].

2.2 | Data Preparation and Collection

A comprehensive dataset comprising 141 small molecules (cf. Table 1) and 79 polymers (cf. Table 2), each with at least three different polymerization degrees, was constructed to capture a broad range of chemical structures. The small-molecule subset comprises hydrocarbons, alcohols, acids, amines, ketones, aldehydes, nitriles, organic chlorides, and benzene derivatives (Table 1). Figure 1 illustrates the largest representative molecule from each small molecule type.

For polymers, the ML models used features derived from small molecules (repeating elements, RE) that closely mirror the repeating units of polymers, as described previously [25]. For a given polymer, multiple small molecules—including the monomer—may serve as potential RE. Taking polylactic acid (PLA, Figure 2) as an example ($\delta = 19.58$ $MPa^{1/2}$), candidate RE molecules include hydroxyacetone ($\delta = 27.35$ $MPa^{1/2}$), methyl acetate ($\delta = 18.74$ $MPa^{1/2}$), and propionic acid ($\delta = 26.21$ $MPa^{1/2}$). Methyl acetate is ultimately chosen because its δ value aligns most closely with that of PLA, whereas hydroxyacetone and propionic acid—both capable of extensive hydrogen bonding—would be less accurate representations of the PLA repeating unit. The RE structure of each polymer is then encoded as a string using SMILES [61]. Although several polymer-representation schemes have been proposed—including the Hierarchical Editing Language for Macromolecules (HELM) [62], the CurlySMILES notation [63], the Big-SMILES formalism [64], and graph-based molecular ensemble representations [65]—these formats are not directly compatible with descriptor-generation tools such as PaDEL. Therefore, we employed the RE-SMILES representation, which enables consistent and reliable calculation of polymer descriptors for our ML models.

After finalizing all REs and generating the corresponding SMILES, the polymer dataset was compiled. Table 2 summarizes the polymers along with their REs and chain lengths. The polymer dataset is composed mainly of homopolymers, as their REs can be represented in a clear and consistent RE-SMILES

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Summary of the 141 small molecules included in the dataset.

Type	Formula	Size (n)	Number of molecules
hydrocarbons	C_nH_{2n+2}	6-10	25
acids	$C_nH_{2n+1}COOH$	1-6	7
alcohols	$C_nH_{2n+1}OH$	1-10	42
ketones	$C_nH_{2n}O$	3-8	6
amines	$C_nH_{2n+1}NH_2$	1-6	5
aldehydes	$C_{n\text{-}1}H_{2n\text{-}1}CHO$	4-7	17
nitriles	$C_nH_{2n+1}CN$	2-6	4
chlorides	$C_nH_{2n+1}Cl$	1-6	5
aromatics	$C_6H_6/C_6H_5CH_2OCH_3/C_6H_5C_nH_{2n+1}$	6-13	30

form. In addition to homopolymers, we also include alternating copolymers (denoted by "–alt–"), for which the chain sequence is periodic (…ABAB…) and can be represented by a single constitutional repeat unit of the form –A–B–. In these cases, the RE corresponds to the –A–B– repeat unit (containing both monomer-derived units A and B), enabling descriptor generation in the same RE-based format used for homopolymers. In contrast, many copolymer classes (e.g., random, block, graft, or compositionally variable copolymers) require explicit information about sequence, composition, and architecture that is not encoded in the present RE-based representation and are therefore outside the scope of this framework.

2.3 | MD Simulations

The solubility parameters for all small molecules and polymers were determined through a systematic atomistic MD simulation workflow, as illustrated in Figure 3. Initial amorphous models were generated using the Amorphous Cell module, employing a conformational Monte Carlo procedure based on the Theodorou-Suter algorithm [66]. Small molecule systems contained 370–500 molecules per simulation cell, while polymer systems contained 10–60 chains per cell, depending on polymer chain length (cf. Table 2). For each system, 10–20 independent initial configurations were generated to ensure adequate sampling of conformational space, from which the 5–10 lowest-energy structures after geometry optimization were selected for subsequent equilibration.

The equilibration process employed a multi-stage approach. Selected structures first underwent simulated annealing to overcome local energy barriers and identify more stable conformations. This process began with initial equilibration at 298 K for 95 ps using the canonical (NVT) ensemble, followed by stepwise heating from 298 to 998 K in 100 K increments over 7 steps. The temperature was then decreased from 998 K back to 298 K following the reverse protocol. Each temperature step involved 10 ps of equilibration using a 0.5 fs timestep with a velocity-rescaling thermostat [67]. Following annealing, systems were again equilibrated for 400 ps under isothermal-isobaric (NPT) conditions at 298 K and zero pressure. The NPT equilibration consisted of an initial 100 ps period using the Berendsen barostat [68] for rapid convergence, followed by 300 ps using the Parrinello-Rahman barostat [69] for accurate pressure coupling. Temperature control was maintained using the Nosé-Hoover thermostat [70, 71] with a 1 fs integration timestep throughout the NPT simulations.

After equilibration, each system was rescaled to its average cell parameters obtained from the NPT production phase. CED and other thermodynamic properties were then calculated from NVT production simulations for 250 ps at 298 K, with the Hildebrand solubility parameter computed using $\delta = \sqrt{\text{CED}}$. Additional simulation parameters and detailed procedures are provided in the Supporting Information.

2.4 | Descriptors

2.4.1 | PaDEL Descriptors

The PaDEL-Descriptor software [72] was used to calculate 1D and 2D molecular descriptors from the SMILES strings of small molecules and polymer REs. PaDEL generated 1444 descriptors spanning geometric, informational, charge-based, and functional group features, as well as broad molecular properties. Additional details regarding these descriptor types and their interpretations are provided in the Supporting Information.

2.4.2 | Quantum Chemical Descriptors

Quantum chemical descriptors were calculated for the small molecules and polymer REs via DFT using the Becke 3-parameter Lee–Yang–Parr (B3LYP) [73], exchange-correlation functional, the def2-TZVP basis set [74], and Grimme's D3 dispersion correction [75]. Convergence thresholds were set at 10^{-6} Hartree for energy change and 10^{-3} Hartree/Bohr for the gradient norm. Two quantum chemical properties, atomization energy (AE) and quadrupole moment (QM), were computed. The quadrupole moment was defined as the second moment of charge [76]:

$$QM = \frac{1}{3}\left(Q_{xx} + Q_{yy} + Q_{zz}\right) \tag{4}$$

where Q_{xx}, Q_{yy} and Q_{zz} are the diagonal elements of the second moment of the charge tensor.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Overview of the polymers and their respective repeating elements used to parameterize ML models.

Polymer	RE	Chain length
polyethylene glycol	dimethyl ether	40, 50, 60
polylactic acid	methyl acetate	20, 35, 50
polystyrene	ethylbenzene	30, 45, 50
polyvinyl ethyl ether	diethylether	40, 50, 60
polyacrylamide	propanamide	40, 50, 60
polyacrylic acid	propanoic acid	40, 50, 60
polybutylene	butane	40, 50, 60
polyacrylonitrile	propanenitrile	20, 35, 50
polyallyl cyanide	butyronitrile	50, 55, 65
polyallyl acetate	propyl acetate	50, 55, 65
polypropylene	propane	15, 30, 50, 70, 75, 80
polyvinyl acetate	ethyl acetate	40, 50, 60
polymethacrylonitrile	isobutyronitrile	45, 50, 80
polyvinylpyrrolidone	1-ethyl-2-pyrrolidinone	50, 55, 65
polymethyl cyanoacrylate	methyl_2-cyano-2-methylacetate	35, 50, 85
polyvinyl butyrate	ethyl butyrate	40, 50, 95
polyethylene terephthalate	4-(ethoxy carbonyl)benzoic acid	20, 30, 50
polycaprolactone	6-hydroxyhexanal	30, 50, 60
polyethylene	ethane	20, 50, 90
cis-1,4-polyisoprene	2-methyl-2-butene	20, 40, 50
polyvinyl alcohol	ethanol	15,30,50,75
polyvinyl butyral	4-methyl-2-propyl-1,3-dioxane	15, 35, 50
polyvinyl chloride	chloroethane	20, 35, 50
polychloral	trichloroethanol	10, 40, 70
polyoxymethylene	methanol	25, 40, 50
polyvinyl fluoride	fluoroethane	10, 50, 90
polychloroprene	2-chloro-2-butene	10, 30, 45
polyacetylene	ethylene	25, 40, 50
polyepichlorohydrin	3-chloro-1-propanol	40, 50, 60
polyvinyl ether	ethyl methyl ether	25, 40, 50
polynorbornene	vinylcyclopentane	10, 30, 50
polyethylenimine	ethylamine	15, 30, 75
poly(2-hydroxyethyl methacrylate)	2-hydroxyethyl isobutyrate	15, 30
polyglycolic acid	glycolaldehyde	15, 40, 80
poly(n-isopropylacrylamide)	N-isopropyl	15, 30, 75
polyamide 6	6-aminohexanal	10
polyamide 66	N-(5-aminopentyl)-6-oxoheptanamide	10
polybutylene succinate	4-hydroxybutyl_4-oxobutanoate	10
polycarbonate	formic acid 4-[1-methyl-1-(4-hydroxyphenyl) ethyl] phenyl ester	10, 20, 30
polyether ether ketone	4-(4-phenoxybenzoyl) phenol	10
polyurethane	2-hydroxyethyl_n-[4-[(4-formamidophenyl) methyl] phenyl] carbamate	10

(Continues)

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | (Continued)

Polymer	RE	Chain length
polyphtalamide	N-(aminomethyl)-4-formylbenzamide	10, 15, 20
polyphenylenether	2,6-dimethylphenol	10
poly(2-ethyl-2-oxazolin)	N-ethylpropionamide	10
poly(2-methyl-2-oxazolin)	N-ethylacetamide	10
poly(2-propyl-2-oxazolin)	N-ethylbutanamide	10
polypeptoid α	2-methylaminoacetaldehyde	10
polypeptoid β	3-methylaminopropionaldehyde	10
polypeptoid γ	4-(methylamino)butanal	10
polymethylpentene	2-methylpentane	10, 30, 45
poly ether ketone ketone	4-(4-phenoxybenzoyl) benzaldehyde	10
polyketone	propanaldehyde	10, 30, 50
polypropylene carbonate	carbonic acid isopropyl ester	10
polybutylene terephthalate	4-hydroxybutyl_4-formylbenzoate	10
Polycyclohexylenedimethylene terephthalate	4-formylbenzoic_acid_4-(hydroxymethyl)cyclohexylmethyl_ester	10
polytrimethylene terephthalate	terephthalic acid monopropyl ester	10, 20, 30
polyethylene naphthalate	naphthalene-2,6-dicarboxylic_acid_2-ethyl_ester	10
poly(3-hydroxypropionate)	3-hydroxypropanal	10
poly(3-hydroxybutyrate)	3-hydroxybutanal	10
poly(3-hydroxyvalerate)	3-hydroxypentanal	10
poly(3-hydroxyhexanoate)	3-hydroxyhexanal	10
poly(3-hydroxyheptanoate)	3-hydroxyheptanal	10
poly(3-hydroxyoctanoate)	3-hydroxyoctanal	10
poly(3-hydroxynonaoate)	3-hydroxynonanal	10
poly(3-hydroxydecanoate)	3-hydroxydecanal	10
poly(3-hydroxyundecanoate)	3-hydroxyundecanal	10
poly(3-hydroxydodecanoate)	3-hydroxydodecanal	10
poly(3-hydroxytetradecanoate)	3-hydroxytetradecanal	10
epichlorohydrin rubber	1-chloro-2-propanol	50
poly(4-vinylphenol)	4-ethylphenol	50, 55, 65
poly{succinic acid-alt-[bis(2-oxazoline)]}	4-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-4-oxobutanoic acid	10
poly{adipic acid-alt-[bis(2-oxazoline)]}	6-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-6-oxohexanoic acid	10
poly{dodecanedioic acid-alt-[bis(2-oxazoline)]}	12-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-12-oxododecanoic acid	10
poly{dimethylglutaric acid-alt-[bis(3-oxazoline)]}	5-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-3,3-dimethyl-5-oxopentanoic acid	10
poly{diethylglutaric acid-alt-[bis(2-oxazoline)]}	2-ethyl-4-((2-(2-(ethylamino)-2-oxoacetamido)ethoxy)carbonyl)hexanoic acid	10

(Continues)

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | (Continued)

Polymer	RE	Chain length
poly{diethylmalonic acid-alt-[bis(3-oxazoline)]}	2-ethyl-2-((2-(2-(ethylamino)-2-oxoacetamido)ethoxy)carbonyl)butanoic acid	10
poly{cyclopentanediacetic acid-alt-[bis(2-oxazoline)]}	2-(1-(2-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-2-oxoethyl)cyclopentyl)acetic acid	10
poly{cyclohexanediacetic acid-alt-[bis(2-oxazoline)]}	2-(1-(2-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-2-oxoethyl)cyclohexyl)acetic acid	10
poly{phenylsuccinic acid-alt-[bis(2-oxazoline)]}	4-(2-(2-(ethylamino)-2-oxoacetamido)ethoxy)-4-oxo-2-phenylbutanoic acid	10

2.4.3 | Descriptors Related to Polymer Chain and Local Environment

Descriptors specifically related to the polymer chain include the molecular weight and the number of repeating units (chain length). Additionally, local structural descriptors, such as zero- and first-order connectivity indices, were calculated following Bicerano's approach [77]. Two atomic indices, δ and δ^{v}, characterize the bonding and electronic environment of each non-hydrogen atom. The index δ is a simple connectivity measure denoting the number of non-hydrogen atoms bonded to a given atom, whereas δ^{v} incorporates the electron configuration of an atom and is defined by

$$\delta^{v} = \frac{z^{\nu} - N_{\mathrm{H}}}{z - z^{\nu} - 1} \tag{5}$$

where z^{ν} is the number of valence electrons, N_{H} is the number of hydrogens bound to it, and z is its atomic number [78]. The zeroth-order (atomic) connectivity indices ${}^{0}\chi^{1}$ and ${}^{0}\chi^{1}$ are defined in terms of the sums

$${}^{0}\chi = \sum \frac{1}{\sqrt{\delta}} \tag{6}$$

and

$${}^{0}\chi^{1} = \sum \frac{1}{\sqrt{\delta^{v}}} \tag{7}$$

The connectivity indices were calculated using the Synthia module in Materials Studio [79]. Combined with descriptors from PaDEL and quantum chemical calculations, these chain-related features yielded a total of 1449 descriptors in the final dataset.

2.5 | Data Cleaning, Descriptors Selection, and Learning

The data collection phase involved preprocessing steps to eliminate missing or anomalous values and to remove descriptors containing only zeros. The resulting dataset, comprising 1083 descriptors, was then divided into training and testing subsets in an 80:20 ratio and subsequently standardized. Descriptor selection techniques, including PCC, PCA, and RFE were applied to identify the most relevant features. The selected descriptors were used to train ML algorithms that included SVR, RF, LASSO, GPR, and NNs. All models, except GPR, were trained using a systematic fivefold cross-validation scheme [80], and a grid search for hyperparameter optimization on the training data to avoid data leakage and reduce variance. In this process, five-fold cross-validation was applied with the coefficient of determination (R^2) as the scoring metric. For each hyperparameter combination, the mean R^2 across the five validation folds was computed, and the combination with the highest mean R^2 was selected. The model was then refitted on the full training set using these optimal hyperparameters. GPR was not subjected to cross-validation and grid search, as it internally optimizes kernel parameters by maximizing the log marginal likelihood during training [81]. The final refitted model was used to generate predictions for both the training and test datasets. Model performance was then quantitatively assessed by comparing the predicted values with the corresponding true values in each dataset. The evaluation metrics included the coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE), ultimately yielding the final optimized model. Detailed workflow information is available in Figure S1.

For SVR, both linear and radial basis function (RBF) kernels were tested, with the penalty parameter C and the γ parameter (for the RBF kernel) controlling the influence of individual training samples. RF was tuned by adjusting the number of trees ($n_{\mathrm{estimators}}$) and the maximum tree depth (max_{depth}) to balance predictive accuracy and computational efficiency. LASSO was applied with the α parameter to control the extent of shrinkage during feature selection and regularization. The GPR model employed a combination of the Matern kernel and the WhiteKernel to model smoothness and noise, respectively. A feedforward neural network (multilayer perceptron, MLP) was implemented with two hidden layers, the Rectified Linear Unit (ReLU) activation function, and the Adam optimizer.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

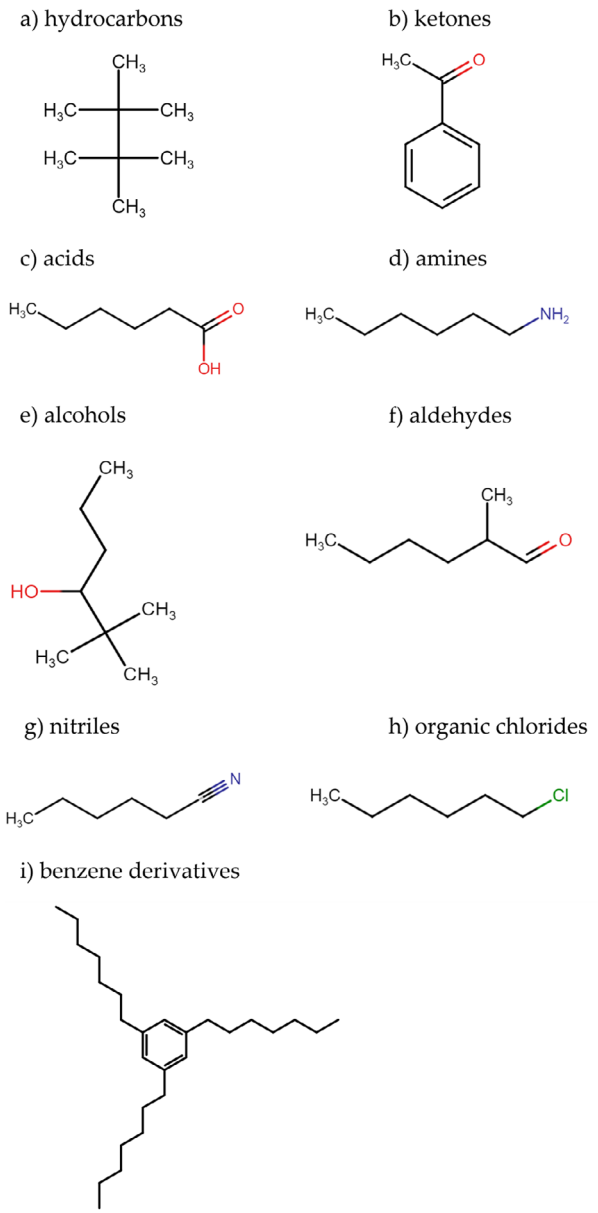

FIGURE 1 | The largest molecule of each small molecule type included in the dataset: (a) 2,2,3,3 tetramethyl butane (C_8H_{18}), (b) acetophenone (C_8H_8O), (c) hexanoic acid ($C_6H_{12}O_2$), (d) hexylamine ($C_6H_{15}N$), (e) 2,2-dimethyl-3-hexanol ($C_8H_{18}O$), (f) 2-methyl hexanal ($C_7H_{14}O$), (g) hexane nitrile ($C_6H_{11}N$), (h) chlorohexane ($C_6H_{13}Cl$), and (i) 1,3,5-triheptyl-benzene ($C_{27}H_{48}$).

2.6 | Group Contribution Method

The GC approach, originally developed by Joback and Reid [82], was applied to small molecules to calculate SP. The GC-based enthalpy of vaporization is given as

$$\Delta H_{\text{vap}} = 15.30 + \sum_{\text{i}} n_i (\Delta H_{\text{vb}})_i \tag{8}$$

where n_i is the number of the group contribution of type i, and $(\Delta H_{\text{vb}})_i$ is the group contribution of type i. The group contribution values $(\Delta H_{\text{vb}})_i$ were taken from the Joback and Reid method [83].

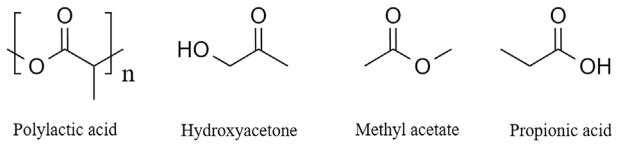

FIGURE 2 | PLA and its potential RE candidates: hydroxyacetone, methyl acetate, and propionic acid.

This enthalpy of vaporization was then used to calculate the SP according to Equation (1).

3 | Results and Discussion

3.1 | Degree of Polymerization and Number of Polymer Chains

Polymer chain length is known to influence polymer properties and results of MD simulations [84]. As shown in Figure 4, for polyethylenimine (PEI), the SP decreases significantly from 22.22 $MPa^{1/2}$ to 18.59 $MPa^{1/2}$ as the number of repeating units increases from 15 to 450, indicating a strong dependence of SP on chain length.

We further determined that the number of polymer chains in the simulation cell had a negligible effect on the SP. For instance, in the case of PEI, as the number of polymer chains increased from 10 to 30, the SP remained nearly unchanged, varying only slightly from 22.16 $MPa^{1/2}$ to 22.22 $MPa^{1/2}$. Similar trends were observed for poly(N-isopropylacrylamide) (PNIPA) (17.69–17.88) $MPa^{1/2}$, poly(2-hydroxyethyl methacrylate) pHEMA (20.22–20.26) $MPa^{1/2}$, and polyglycolic acid PGA (25.48–25.39) $MPa^{1/2}$ over the same range. Consequently, the number of polymer chains in the simulation cell was not considered an effective descriptor for the MD simulations. Instead, smaller simulation cells, containing 10 polymer chains, were used to reduce computational cost. A detailed graphical representation of these results is provided in Figure S2.

3.2 | Descriptor Selection

Robust descriptor selection is critical for developing accurate and generalizable machine learning models, especially when dealing with large, chemically diverse datasets. In this work, we systematically reduced the dimensionality of our feature space and identified the most informative descriptors for solubility parameter prediction by combining several complementary techniques.

We began by evaluating the linear correlation between each descriptor and the target SP using PCC. Only descriptors exhibiting moderate or strong correlation (PCC > 0.5) were retained, narrowing the set to 43 features and eliminating those with weak or redundant relationships to SP (see Figure S3). To further address multicollinearity and reduce feature dimensionality, we also performed PCA. The first principal component accounted for 33% of the total variance, and 44 components were ultimately selected to capture 95% of the variance in the dataset. The optimal number of principal components was determined by incorpo-

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

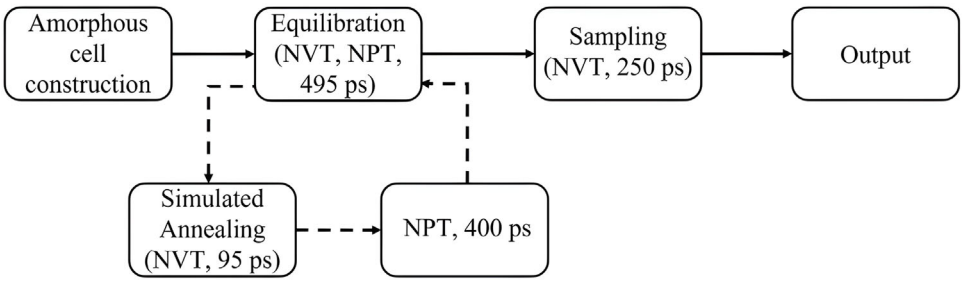

FIGURE 3 | Overview of the MD simulation workflow for SP calculation, including amorphous cell construction, equilibration, and sampling steps with key simulation phases (NPT and NVT) and simulation times in ps.

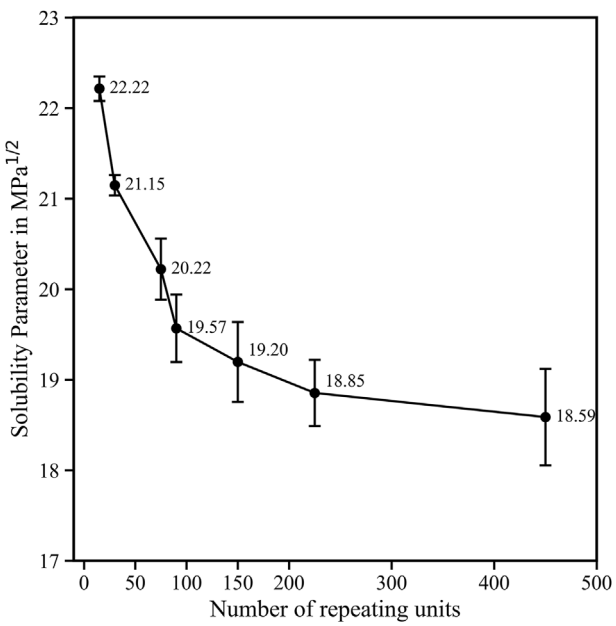

FIGURE 4 | SP values derived from atomistic MD simulations for polyethylenimine (PEI) with different chain lengths. The vertical lines at each point represent error bars corresponding to the standard deviation from multiple simulation samples.

rating grid search during model hyperparameter optimization, ensuring that dimensionality reduction did not compromise predictive performance (see Figure S4).

For model-specific feature refinement, RFE was applied, particularly with the SVR model. RFE systematically removes the least relevant features, iteratively optimizing model performance and resulting in an optimal subset of 50 descriptors for SVR (Table S3). For the LASSO and RF models, RFE with cross-validation (RFECV) was used to select features in a fully data-driven manner, yielding 216 descriptors for LASSO and 14 descriptors for RF (final RF set in Table S5).

By integrating PCC-based filtering, unsupervised dimensionality reduction via PCA, and model-driven pruning through RFE/RFECV, we identified compact, information-rich descriptor sets tailored to each learning algorithm. This rigorous selection strategy not only minimized overfitting and reduced computational burden but also ensured that the retained features captured the underlying chemical and structural determinants of the solubility parameter, enabling accurate, interpretable, and transferable predictions across both polymers and small molecules.

3.3 | Hyperparameters Optimization

Twelve ML models were constructed by combining four ML algorithms (SVR, LASSO, RF, and GPR) and three feature engineering techniques (PCC, PCA, and RFE). Each model is denoted by the notation Model/Reduction, where Model represents the prediction algorithm and Reduction denotes the feature engineering technique. SVR/PCC combines PCC-based feature selection with SVR; the same pairing yields LASSO/PCC, RF/PCC, and GPR/PCC, where PCC-selected features are learned by LASSO, RF, and GPR, respectively. The PCA pipelines project the original descriptors onto principal components, then train SVR, LASSO, RF, and GPR on those components. This produces SVR/PCA, LASSO/PCA, RF/PCA, and GPR/PCA. Recursive selection uses SVR/RFE, where RFE is coupled with SVR. LASSO/RFECV, RF/RFECV, and GPR/RFECV pair RFECV-refined features with LASSO, RF, and GPR, respectively. Detailed results of the optimized hyperparameters of the ML models are provided in the Supporting Information (Table S7). In addition to these ML models, NN-based models were also implemented to evaluate the effectiveness of different feature selection methods in predicting SP. Specifically, Neural Network with PCC (NN/PCC), Neural Network with PCA (NN/PCA), and Neural Network with RFECV (NN/RFECV) were used. Optimized hyperparameters for each NN model are provided in Supporting Information (Table S8).

The average training times for the models varied significantly depending on the feature selection strategy used. Each combination of the feature selection method and model architecture affects the computational complexity and thus influences training duration. All computations were performed on a Windows 11 system equipped with an Intel Core i9-14900T CPU (1.10 GHz, 24 cores) and 64 GB RAM. According to our setup, training was carried out using a single CPU core. Among all models, GPR consistently showed the fastest training time (0.05 s) across all feature selection techniques, as it was not subjected to cross-validation or grid search (as explained in Section 2.5). Among the models that used cross-validation and grid search, LASSO/PCA completed training in the shortest time (0.08 s), followed by LASSO/PCC (0.66 s). In contrast, the RF/RFECV model required the longest training time (11.47 min), followed by RF/PCA (11.25 min). All NN models demonstrated high computational efficiency, with

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Comparison of machine learning model performance in predicting solubility parameters.

Model	Training			Test		
	R^2	MAE[a)]	RMSE[a)]	R^2	MAE[a)]	RMSE[a)]
SVR/PCC	0.86	1.01	1.57	0.75	1.26	1.88
RF/PCC	0.95	0.45	0.97	0.90	0.84	1.17
LASSO/PCC	0.87	1.19	1.52	0.78	1.33	1.76
GPR/PCC	0.97	0.43	0.69	0.92	0.73	1.10
SVR/PCA	0.90	0.41	1.37	0.87	0.88	1.35
RF/PCA	0.94	0.52	1.07	0.80	1.09	1.68
LASSO/PCA	0.85	1.24	1.64	0.77	1.38	1.81
GPR/PCA	0.98	0.40	0.66	0.88	0.84	1.30
SVR/RFE	0.95	0.51	0.94	0.83	1.01	1.58
RF/RFECV	0.94	0.42	1.01	0.93	0.68	0.99
LASSO/RFECV	0.97	0.44	0.68	0.90	0.76	1.21
GPR/RFECV	0.99	0.21	0.41	0.94	0.54	0.96

[a)] $MPa^{1/2}$.

training times of less than one minute for each model on the same computing platform as the ML models. Detailed information regarding the training times for all models is provided in Table S9.

3.4 | Performance of ML Models

The detailed performance metrics (R^2, MAE, and RMSE) for both training and test sets are summarized in Table 3, providing a clear comparison of model accuracy and generalization across various feature selection methods and algorithms.

The predictive performance of the ML models shows clear differences in both accuracy and generalization across feature-selection strategies and algorithms (cf. Table 3). The best overall results are obtained with GPR/RFECV, which reaches R^2_{test} = 0.94 with MAE_{test} = 0.54 and $RMSE_{test}$ =0.96. RF/RFECV follows closely with R^2_{test} =0.93, MAE_{test} = 0.68, and $RMSE_{test}$ =0.99, and it exhibits a small train—test gap in R^2 (0.94 vs 0.93), indicating stable generalization. Using simple correlation filtering, GPR/PCC and RF/PCC also perform well, with R^2_{test} = 0.92 and 0.90, respectively. PCA helps SVR relative to PCC, as SVR/PCA improves to R^2_{test} =0.87 from 0.75 for SVR/PCC, although RF/PCA and GPR/PCA underperform their PCC and RFECV counterparts. SVR/RFE achieves high training accuracy (R^2_{train} =0.95) but drops to R^2_{test} = 0.83, which points to overfitting. Linear LASSO variants are the least accurate, with LASSO/PCC and LASSO/PCA yielding R^2_{test} = 0.78 and 0.77, and the largest test errors among all pipelines. Overall, RFECV consistently improves test performance for non-linear models compared with PCC and PCA, suggesting that task-specific feature subsets are more effective than global variance projections. For practical use, GPR/RFECV offers the highest accuracy, while RF/RFECV delivers nearly the same generalization with modest computational cost. LASSO remains useful for feature screening, but its predictive power is limited on this dataset relative to GPR, RF, and SVR.

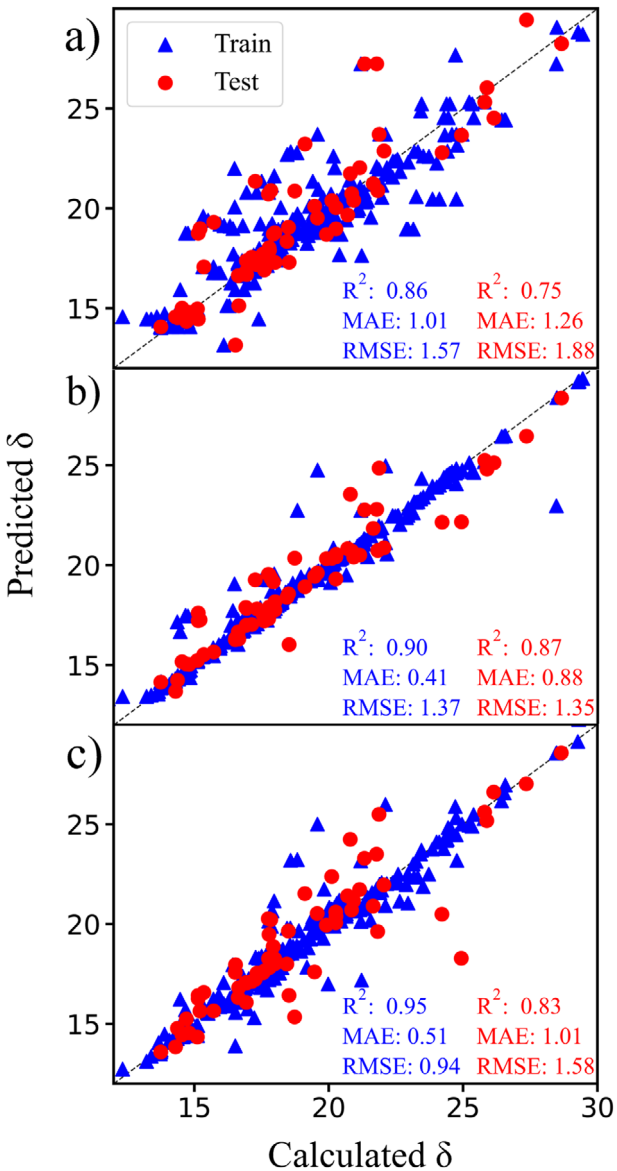

FIGURE 5 | Performance of SVR model with different feature selection methods: (a) PCC, (b) PCA, and (c) RFE. MAE and RMSE are in $MPa^{1/2}$.

3.4.1 | SVR Model Performance

Figure 5 compares SVR models built with three feature-selection strategies. Among these variants, SVR/PCA delivered the strongest and most consistent generalization. On the test set, it achieved MAE_{test} =0.88 and $RMSE_{test}$ =1.35 $MPa^{1/2}$, outperforming SVR/RFE (MAE_{test} =1.01, $RMSE_{test}$ =1.58) and SVR/PCC (MAE_{test} =1.26, $RMSE_{test}$ =1.88). SVR/PCA also balanced fit and generalization best, with R^2_{train} = 0.90 and R^2_{test} = 0.87 for a gap of 0.03. In contrast, SVR/RFE reached a higher training R^2_{train}(0.95) but dropped to R^2_{test} = 0.83, which indicates overfitting. This

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

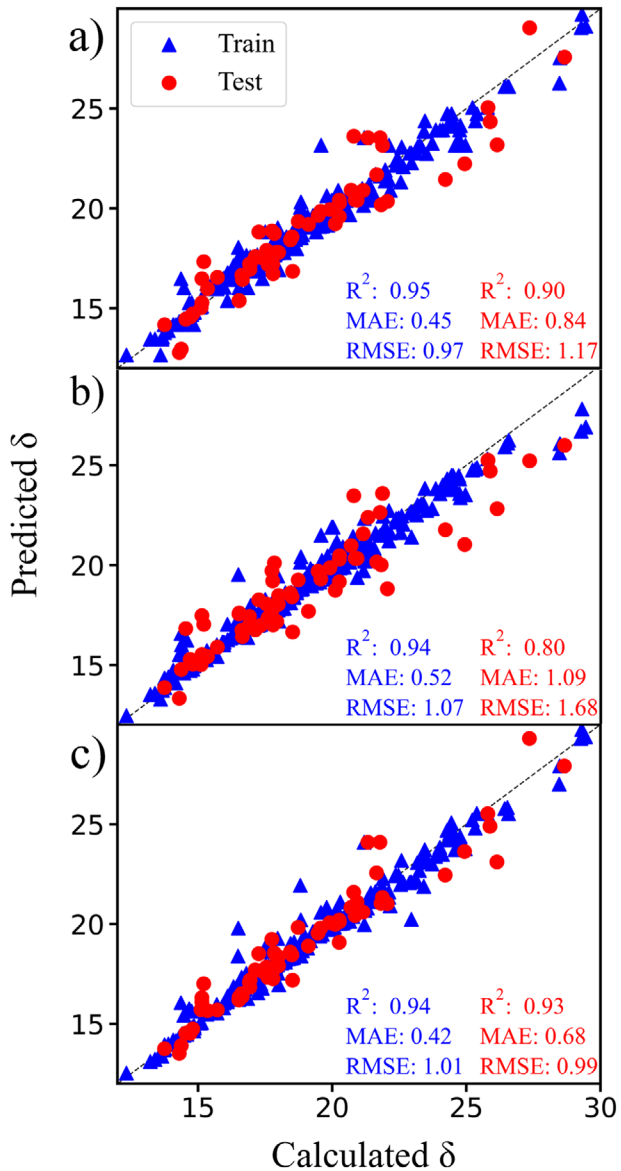

FIGURE 6 | Performance of RF model with different feature selection methods: (a) PCC, (b) PCA, and (c) RFECV. MAE and RMSE are in $MPa^{1/2}$.

suggests that, for the SVR model, representing the descriptor space by principal components that resolve multicollinearity and reduce dimensionality is more effective than either linear correlation filtering (PCC) or RFE-based removal of the least relevant features. Taken together, these results support PCA as the preferred feature strategy for SVR, offering lower error, tighter train–test agreement, and shorter runtime (cf. Table 3 and Figure 5).

3.4.2 | RF Model Performance

Figure 6 summarizes the performance of the RF models across three feature-selection strategies. The most important descriptors identified by each method are listed in the Supporting Information (Table S10). Across variants, RF/RFECV and RF/PCC provide the strongest and most balanced performance on both training and test data. RF/RFECV yields the highest predictive accuracy with R^2_{test} =0.93, MAE_{test} =0.68, and $RMSE_{test}$ =0.99 $MPa^{1/2}$, while maintaining a very small train–test gap in R^2 (0.94 vs 0.93). RF/PCC performs comparably well with R^2_{test} =0.90, MAE_{test} =0.84, and $RMSE_{test}$ =1.17 $MPa^{1/2}$, and it also shows good generalization (R^2_{train} = 0.95 vs R^2_{test} =0.90). RF/PCA underperforms relative to these two alternatives with R^2_{test} = 0.80, MAE_{test} =1.09, and $RMSE_{test}$ =1.68 $MPa^{1/2}$, and it shows a larger train-test gap. These results suggest that RF achieves better performance with descriptor subsets that retain the original feature structure (PCC and RFECV), while PCA transformations may smooth out descriptor-specific patterns important for accurate prediction in tree-based models.

3.4.3 | LASSO Model Performance

Figure 7 compares LASSO models built with three feature-selection strategies. RFECV delivers the highest predictive accuracy, achieving R^2_{test} =0.90, MAE_{test} =0.76, and $RMSE_{test}$ = 1.21 $MPa^{1/2}$. In contrast, correlation filtering and PCA perform markedly worse: LASSO/PCC yields R^2_{test} =0.78 with MAE_{test} =1.33 and $RMSE_{test}$ =1.76 $MPa^{1/2}$, and LASSO/PCA reaches R^2_{test} = 0.77 with MAE_{test} = 1.38 and $RMSE_{test}$ =1.81 $MPa^{1/2}$. RFECV also maintains a balanced fit, with a modest train-test gap relative to the PCC and PCA variants. These results show that the RFECV approach, which progressively discards weak descriptors to refine the feature set, is particularly well aligned with LASSO's least-squares-based estimation and coefficient-shrinkage behavior, providing stronger generalization and substantially lower error.

3.4.4 | GPR Model Performance

Figure 8 compares GPR models built with three feature-selection strategies. RFECV selects a compact subset of 14 descriptors and delivers the strongest predictive performance, with R^2_{test} = 0.94, MAE_{test} =0.54, and $RMSE_{test}$ =0.96 $MPa^{1/2}$. GPR/PCC performs well but trails GPR/RFECV, yielding R^2_{test} = 0.92, MAE_{test} =0.73, and $RMSE_{test}$ =1.10 $MPa^{1/2}$. GPR/PCA underperforms both alternatives with R^2_{test} =0.88, MAE_{test} =0.84, and $RMSE_{test}$ =1.30 $MPa^{1/2}$. GPR/RFECV also shows the smallest train-test R^2 gap among the three variants (0.99 vs 0.94), indicating stable generalization. Overall, these results support RFECV as the preferred feature-selection method for GPR on this dataset, providing superior accuracy and robustness (cf. Table 3 and Figure 8). In addition, compared to SVR, RF, and LASSO, the Bayesian GPR framework shows greater flexibility to different forms of feature engineering, producing consistently accurate results with PCC, PCA, and RFECV.

3.4.5 | NN Model Performance

Figure 9 compares NN models built with three feature-selection strategies. Across variants, NN/PCC delivers the strongest and most consistent generalization, achieving R^2_{test} =0.88 and MAE_{test} =0.92 $MPa^{1/2}$. NN/RFECV and NN/PCA perform less well, with R^2_{test} = 0.83 and 0.77 and MAE_{test} = 1.24 and 1.22 $MPa^{1/2}$,

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

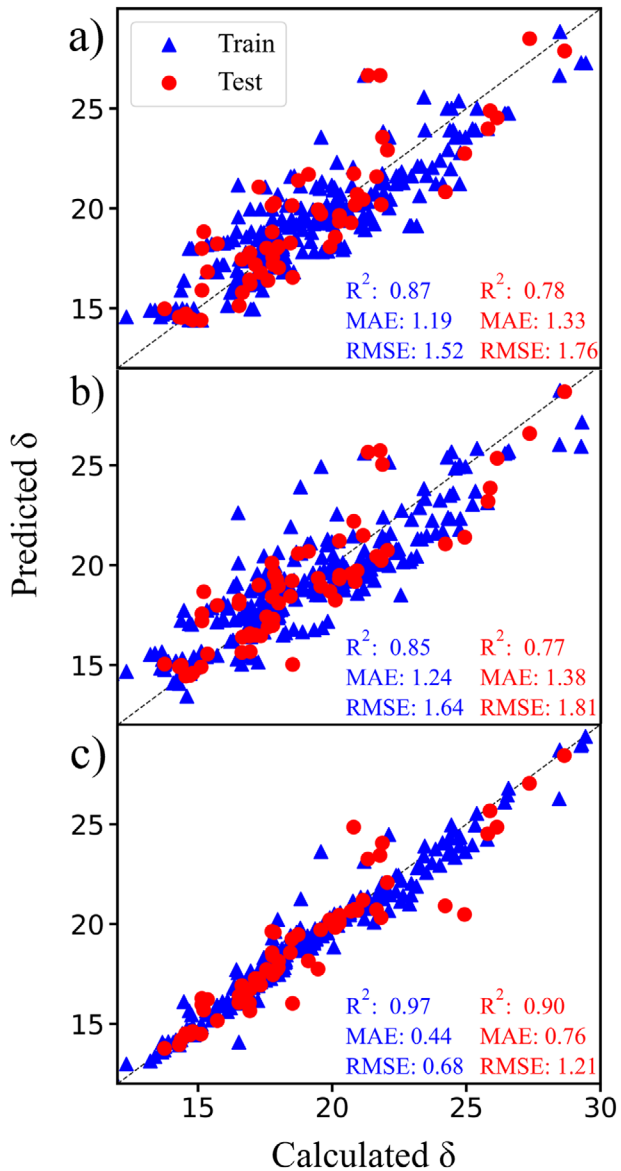

FIGURE 7 | Performance of LASSO model with different feature selection methods: (a) PCC, (b) PCA, and (c) RFECV. MAE and RMSE are in $MPa^{1/2}$.

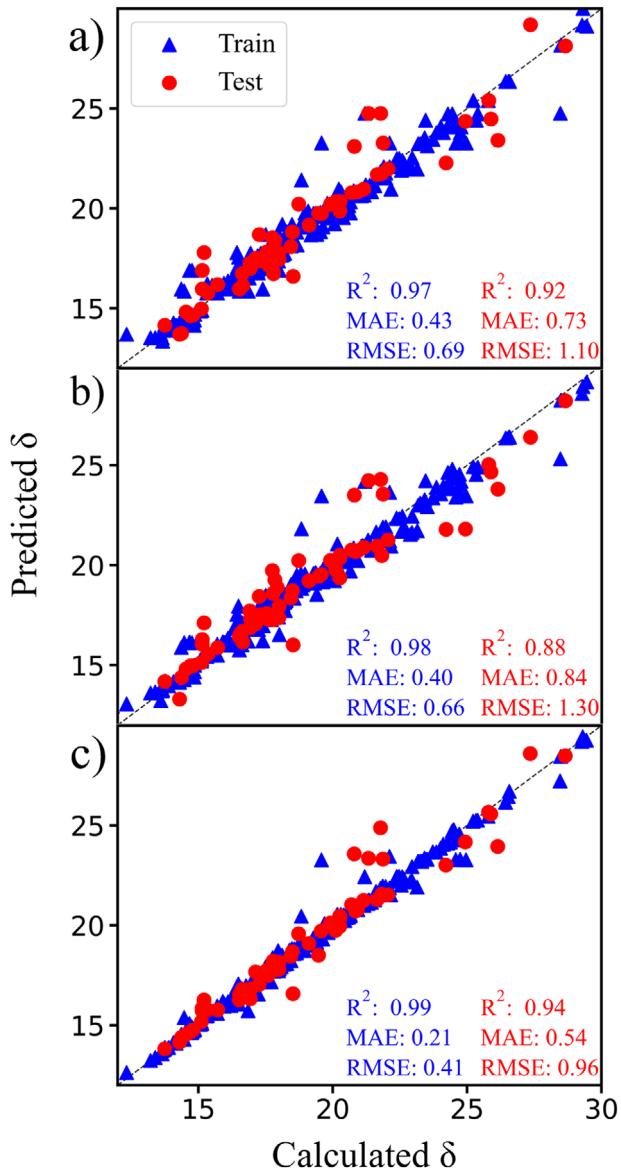

FIGURE 8 | Performance of GPR model with different feature selection methods: (a) PCC, (b) PCA, and (c) RFECV. MAE and RMSE are in $MPa^{1/2}$.

respectively. Overall, the NN models do not surpass the top-performing traditional regressors in this study, but when an NN architecture is desired, PCC emerges as the preferred feature-selection approach due to its lower error and more reliable test performance. Importantly, the limited advantage of NN over ML methods (GPR, RF) aligns with the understanding that neural networks typically require significantly larger datasets to achieve optimal predictive performance.

3.5 | Comparing GC and ML Approaches for SP Prediction

Figure 10 shows the comparison between the GC method and MD results for small molecules. The GC method gives moderate accuracy, with an R^2 of 0.56, MAE of 1.87, and RMSE of 2.19. In contrast, all ML models discussed earlier achieved better performance than the GC method. Even the lowest R^2_{test} among the ML models was 0.75, and the highest MAE_{test} and $RMSE_{test}$ values were 1.38 and 1.88 $MPa^{1/2}$, respectively (c.f. Table 3), still outperforming the GC method. These results highlight that ML models consistently provide more accurate and reliable predictions for SP values compared to the GC method.

Although the GC method is generally viewed as straightforward and practical, Rheineck and Lin [85] have shown that the additivity principle underlying the approach does not strictly hold for cohesive energy calculations at room temperature. They reported that functional group contributions fluctuate depending on the broader molecular environment.

Applying GC methods to polymers presents additional hurdles due to the limited availability of precise, reliable data specific to polymer systems. Bunn's findings indicate that cohesion energy

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

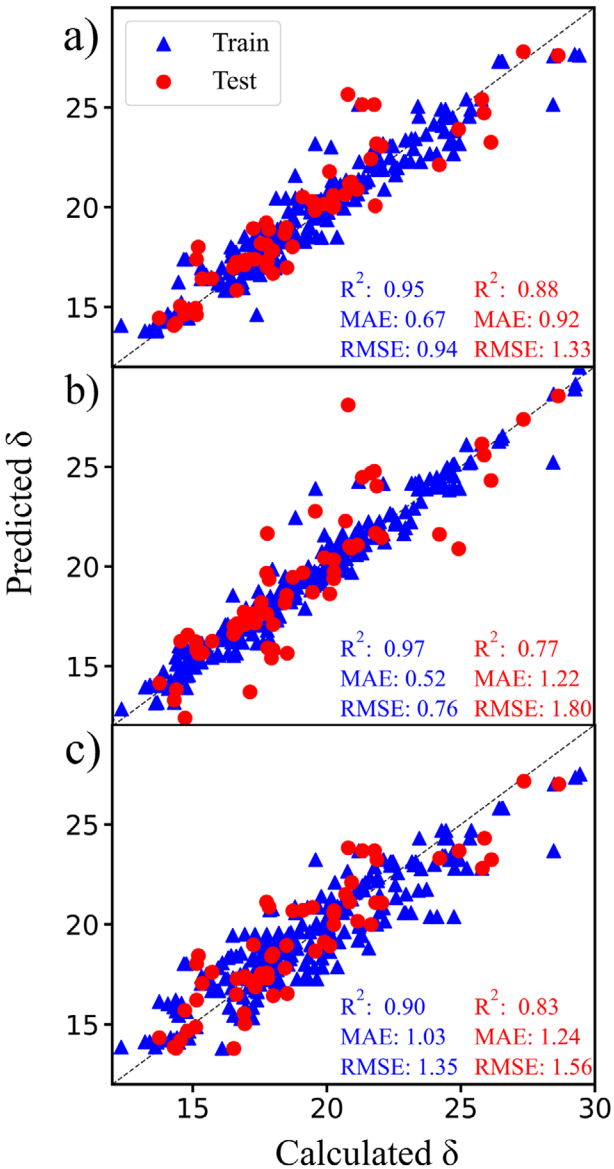

FIGURE 9 | Performance of NN with different feature selection methods: (a) PCC, (b) PCA, and (c) RFECV. MAE and RMSE are in $MPa^{1/2}$.

values derived from boiling-point data are suitable for small molecules but require correction for polymers, as the molecular volumes of polymers are smaller than their additive group contributions, leading to higher cohesion energies [86]. Furthermore, the structural complexity of polymers, along with their varied intermolecular interactions, complicates property estimation by traditional GC methods.

Even though GC approaches can produce approximate estimates for some polymer properties, they do not account for finer structural distinctions, such as those between geometrical isomers or positional substituents. As a result, GC methods may generate identical SP values for polymers with distinct structural and physical characteristics [87]. Addressing these shortcomings demands more advanced techniques that can capture the complexity of polymer systems. In particular, atomistic MD simulations and ML models incorporate detailed structural and interaction-based

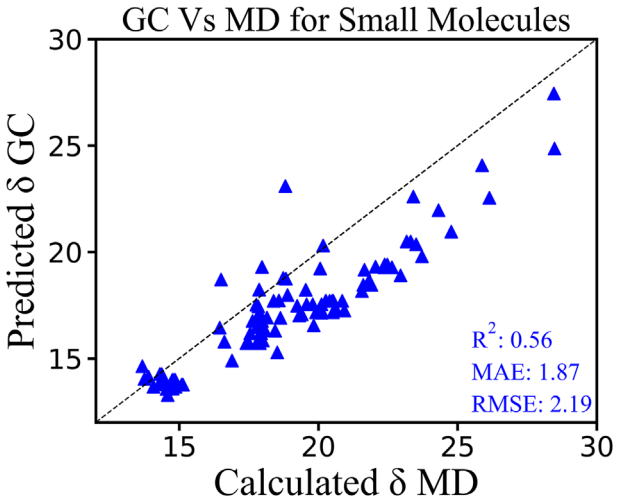

FIGURE 10 | Comparison of GC-predicted SP values with MD calculated values for small molecules. MAE and RMSE are in $MPa^{1/2}$.

information, expecting to yield more accurate polymer property predictions.

4 | Conclusions

We developed and assessed a hybrid route to Hildebrand SP that couples atomistic MD reference data with supervised learning on molecular, quantum-chemical, and chain-aware descriptors. The approach captures key physics of polymer systems, most notably the sensitivity of SP to chain length (e.g., for PEI, the change of SP from 22.22 $MPa^{1/2}$ to 18.59 $MPa^{1/2}$ when increasing from 15 to 450 repeat units), while confirming that the number of chains per simulation cell has negligible impact within the ranges studied. Trained on 79 polymers and 141 small molecules, the resulting models are applicable to new molecules and polymers beyond the training set, providing SP estimates suitable for downstream use in miscibility assessments (e.g., via the Flory-Huggins interaction parameter χ).

A head-to-head comparison of learning algorithms and feature-selection strategies yields clear guidance. GPR performs best and most stably across selection methods, with GPR/RFECV reaching $R^2 = 0.94$, MAE = 0.54, RMSE = 0.96 $MPa^{1/2}$ on the test set. RF delivers similarly strong accuracy, and RF/RFECV achieves $R^2 = 0.93$, but requires higher computation during training. LASSO and SVR are sensitive to feature engineering: PCC and PCA variants show higher test errors, whereas coupling with RFE/RFECV improves generalization and reduces error. Neural networks reach consistent accuracy, though they do not surpass the top GPR/RF pipelines in this study. Across the board, all ML models outperform the conventional group-contribution method for small molecules (GC: $R^2 = 0.56$, MAE = 1.87, RMSE = 2.19 $MPa^{1/2}$), indicating a practical advantage of data-driven approaches over purely additive schemes.

In practice, MD-derived SP values are effective training targets where experiments are scarce. Chain length should be retained explicitly among descriptors, and task-tuned feature subsets identified by RFECV are a robust default. For accuracy with efficiency,

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

GPR/RFECV is a sensible first choice, with RF/RFECV a strong alternative when feature importance profiles are needed. This study treats δ as a single-number descriptor and does not explicitly resolve polar or hydrogen-bonding contributions. Extending the framework to Hansen-type parameters and mixture-aware descriptors, broadening experimental calibration, and incorporating uncertainty estimates for domain-of-applicability checks and active data acquisition are natural next steps. Overall, combining MD labels with curated descriptors and non-linear regression provides an accurate, scalable approach to predicting solubility parameters for molecules and polymers.

Finally, while the present dataset consists primarily of homopolymers and a limited set of alternating copolymers (denoted by "–alt–") that are captured using a single –A–B– repeat unit, a full treatment of other copolymer classes (e.g., random, block, graft, or compositionally variable copolymers) introduces additional complexity arising from sequence effects, blockiness, and branching, which extend beyond the scope of this study. Future work will therefore focus on adapting the framework to more diverse and architecturally complex copolymers. This will require polymer-specific representations capable of encoding hierarchical structural information at the full-polymer, backbone, and side-chain levels, as demonstrated in recent developments such as PolyMeTriX [88]. Incorporating such advanced featurization strategies will enable the accurate modeling of sequence-defined and architecturally complex polymer systems.

Acknowledgements

The work was supported by Deutsche Forschungsgemeinschaft (DFG), German Research Foundation, project number 316213987, SFB 1278, project A01 as well as project number 444711651, GRK 2723, project F.

Funding

Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, project number 316213987, SFB 1278, project A01 as well as project number 444711651, GRK 2723, project F).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.16981636, reference number 16981636.

References

1. B. C. Hancock, P. York, and R. C. Rowe, "The Use of Solubility Parameters in Pharmaceutical Dosage Form Design," *International Journal of Pharmaceutics* 148 (1997): 1–21, https://doi.org/10.1016/S0378-5173(96)04828-4.

2. Y. Luo, X. Chen, S. Wu, S. Cao, Z. Luo, and Y. Shi, "Molecular Dynamics Simulation Study on Two-Component Solubility Parameters of Carbon Nanotubes and Precisely Tailoring the Thermodynamic Compatibility Between Carbon Nanotubes and Polymers," *Langmuir* 36 (2020): 9291–9305, https://doi.org/10.1021/acs.langmuir.0c01736.

3. S. Venkatram, C. Kim, A. Chandrasekaran, and R. Ramprasad, "Critical Assessment of the Hildebrand and Hansen Solubility Parameters for Polymers," *Journal of Chemical Information and Modeling* 59 (2019): 4188–4194, https://doi.org/10.1021/acs.jcim.9b00656.

4. C. Kim, A. Chandrasekaran, T. D. Huan, D. Das, and R. Ramprasad, "Polymer Genome: A Data-Powered Polymer Informatics Platform for Property Predictions," *The Journal of Physical Chemistry C* 122 (2018): 17575–17585, https://pubs.acs.org/doi/10.1021/acs.jpcc.8b02913.

5. A. Erlebach, T. Ott, C. Otzen, et al., "Thermodynamic Compatibility of Actives Encapsulated into PEG-PLA Nanoparticles: In Silic o Predictions and Experimental Verification," *Journal of Computational Chemistry* 37 (2016): 2220–2227, https://doi.org/10.1002/jcc.24449.

6. M. Behnke, A. Vollrath, P. Dahlke, et al., "PEtOxylated Polyesteramide Nanoparticles for the Delivery of Anti-Inflammatory Drugs," *Materials Today Chemistry* 35 (2024): 101848, https://doi.org/10.1016/j.mtchem.2023.101848.

7. I. Muljajew, M. Chi, A. Vollrath, et al., "A Combined Experimental and in Silico Approach to Determine the Compatibility of Poly(ester amide)s and Indomethacin in Polymer Nanoparticles," *European Polymer Journal* 156 (2021): 110606, https://doi.org/10.1016/j.eurpolymj.2021.110606.

8. J. Westhoff, C. Weber, V. Bachmann, et al., "Amino Acid-equipped Poly(ester amide) Nanoparticles for Improved Encapsulation Efficiency of Lipophilic Anti-inflammatory 6BIGOE Based on Molecular Dynamic Simulations," *International Journal of Pharmaceutics* 684 (2025): 126182, https://doi.org/10.1016/j.ijpharm.2025.126182.

9. J. Zhu, R. Balieu, and H. Wang, "The Use of Solubility Parameters and Free Energy Theory for Phase Behaviour of Polymer-modified Bitumen: A Review," *Road Materials and Pavement Design* 22 (2021): 757–778, https://doi.org/10.1080/14680629.2019.1645725.

10. M. Rubinstein and R. H. Colby, *Polymer Physics*, Oxford University Press (2003): pp. 137–145, https://doi.org/10.1093/oso/9780198520597.001.0001.

11. L. C. Klepsch, P. Dahlke, M. Behnke, et al., "Matching Drug and Polymer for Efficient Delivery of Anti-inflammatory Drugs: PLGA, Polyesteramides, and Acetalated Dextran," *Journal of Materials Chemistry B* 13 (2025): 15516–15529, https://doi.org/10.1039/D5TB01949D.

12. T. Lindvig, M. L. Michelsen, and G. M. Kontogeorgis, "A Flory–Huggins model based on the Hansen solubility parameters," *Fluid Phase Equilibria* 203 (2002): 247–260, https://doi.org/10.1016/S0378-3812(02)00184-X.

13. D. Pospiech, A. Gottwald, D. Jehnichen, et al., "Determination of Interaction Parameters of Block Copolymers Containing Aromatic Polyesters From Solubility Parameters Obtained From Solution Viscosities," *Colloid & Polymer Science* 280 (2002): 1027–1037, https://doi.org/10.1007/s00396-002-0728-6.

14. S. Carvalho, E. Lucas, G. Gonzalez, and L. Spinelli, "Determining Hildebrand Solubility Parameter by Ultraviolet Spectroscopy and Microcalorimetry," *Journal of the Brazilian Chemical Society* 24 (2013): 130250, https://doi.org/10.5935/0103-5053.20130250.

15. A. Forster, J. Hempenstall, I. Tucker, and T. Rades, "Selection of Excipients for Melt Extrusion with Two Poorly Water-soluble Drugs by Solubility Parameter Calculation and Thermal Analysis," *International Journal of Pharmaceutics* 226 (2001): 147–161, https://doi.org/10.1016/S0378-5173(01)00801-8.

16. L. Constantinou and R. Gani, "New Group Contribution Method for Estimating Properties of Pure Compounds," *AIChE Journal* 40 (1994): 1697–1710, https://doi.org/10.1002/aic.690401011.

17. E. Stefanis, L. Constantinou, and C. Panayiotou, "A Group-Contribution Method for Predicting Pure Component Properties of Biochemical and Safety Interest," *Industrial & Engineering Chemistry Research* 43 (2004): 6253–6261, https://doi.org/10.1021/ie0497184.

18. J. Gupta, C. Nunes, S. Vyas, and S. Jonnalagadda, "Prediction of Solubility Parameters and Miscibility of Pharmaceutical Compounds by Molecular Dynamics Simulations," *The Journal of Physical Chemistry B* 115 (2011): 2014–2023, https://doi.org/10.1021/jp108540n.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

19. M. Belmares, M. Blanco, W. A. Goddard Iii, et al., "Hildebrand and Hansen Solubility Parameters from Molecular Dynamics with Applications to Electronic Nose Polymer Sensors," *Journal of Computational Chemistry* 25 (2004): 1814–1826, https://doi.org/10.1002/jcc.20098.

20. D. P. Faasen, A. Jarray, H. J. W. Zandvliet, E. S. Kooij, and W. Kwiecinski, "Hansen Solubility Parameters Obtained via Molecular Dynamics Simulations as a Route to Predict Siloxane Surfactant Adsorption," *Journal of Colloid and Interface Science* 575 (2020): 326–336, https://doi.org/10.1016/j.jcis.2020.04.070.

21. B. M. Lowe, C. K. Skylaris, N. G. Green, Y. Shibuta, and T. Sakata, "Molecular Dynamics Simulation of Potentiometric Sensor Response: The Effect of Biomolecules, Surface Morphology and Surface Charge," *Nanoscale* 10 (2018): 8650–8666, https://doi.org/10.1039/C8NR00776D.

22. H. Sun, "COMPASS: An Ab Initio Force-Field Optimized for Condensed-Phase Applications Overview with Details on Alkane and Benzene Compounds," *The Journal of Physical Chemistry B* 102 (1998): 7338–7364, https://doi.org/10.1021/jp980939v.

23. F. Cailliez, A. Bourasseau, and P. Pernot, "Calibration of Force-fields for Molecular Simulation: Sequential Design of Computer Experiments for Building Cost-Efficient Kriging Metamodels," *Journal of Computational Chemistry* 35 (2014): 130–149, https://doi.org/10.1002/jcc.23475.

24. D. M. Walden, Y. Bundey, A. Jagarapu, V. Antontsev, K. Chakravarty, and J. Varshney, "Molecular Simulation and Statistical Learning Methods Toward Predicting Drug–Polymer Amorphous Solid Dispersion Miscibility, Stability, and Formulation Design," *Molecules (Basel, Switzerland)* 26 (2021): 182, https://doi.org/10.3390/molecules26010182.

25. M. Chi, R. Gargouri, T. Schrader, K. Damak, R. Maâlej, and M. Sierka, "Atomistic Descriptors for Machine Learning Models of Solubility Parameters for Small Molecules and Polymers," *Polymers* 14 (2022): 26, https://doi.org/10.3390/polym14010026.

26. B. Sanchez-Lengeling, L. M. Roch, J. D. Perea, S. Langner, C. J. Brabec, and A. Aspuru-Guzik, "A Bayesian Approach to Predict Solubility Parameters," *in Advanced Theory and Simulations* 2 (2019), 1800069, https://doi.org/10.1002/adts.201800069.

27. C. H. Chan, M. Sun, and B. Huang, "Application of Machine Learning for Advanced Material Prediction and Design," *EcoMat* 4 (2022): 12194, https://doi.org/10.1002/eom2.12194.

28. J. Nistane, L. Chen, Y. Lee, R. Lively, and R. Ramprasad, "Estimation of the Flory-Huggins Interaction Parameter of Polymer-Solvent Mixtures Using Machine Learning," *MRS Communications* 12 (2022): 1096–1102, https://doi.org/10.1557/s43579-022-00237-x.

29. T.-L. Liu, L.-Y. Liu, F. Ding, and Y.-Q. Li, "A Machine Learning Study of Polymer-Solvent Interactions," *Chinese Journal of Polymer Science* 40 (2022): 834–842, https://doi.org/10.1007/s10118-022-2716-2.

30. Y. Aoki, S. Wu, T. Tsurimoto, et al., "Multitask Machine Learning to Predict Polymer–Solvent Miscibility Using Flory–Huggins Interaction Parameters," *Macromolecules* 56 (2023): 5446–5456, https://doi.org/10.1021/acs.macromol.2c02600.

31. P. Hu, Z. Jiao, Z. Zhang, and Q. Wang, "Development of Solubility Prediction Models with Ensemble Learning," *Industrial & Engineering Chemistry Research* 60 (2021): 11627–11635, https://doi.org/10.1021/acs.iecr.1c02142.

32. W. V. Wilding, R. L. Rowley, and J. L. Oscarson, "DIPPR Project 801 evaluated process design data," *Fluid Phase Equilibria* 150–151 (1998): 413–420, https://doi.org/10.1016/S0378-3812(98)00341-0.

33. J. Ethier, E. R. Antoniuk, and B. Brettmann, "Predicting Polymer Solubility from Phase Diagrams to Compatibility: A Perspective on Challenges and Opportunities," *Soft Matter* 20 (2024): 5652–5669, https://doi.org/10.1039/D4SM00590B.

34. R. S. van den Hurk, B. W. J. Pirok, and T. S. Bos, "The Role of Artificial Intelligence and Machine Learning in Polymer Characterization: Emerging Trends and Perspectives," *Chromatographia* 88 (2025): 357–363, https://doi.org/10.1007/s10337-025-04406-7.

35. A. Chandrasekaran, C. Kim, S. Venkatram, and R. Ramprasad, "A Deep Learning Solvent-Selection Paradigm Powered by a Massive Solvent/Nonsolvent Database for Polymers," *Macromolecules* 53 (2020): 4764–4769, https://pubs.acs.org/doi/10.1021/acs.macromol.0c00251.

36. C. J. Peacock, C. Lamont, D. A. Sheen, V. K. Shen, L. Kreplak, and J. P. Frampton, "Predicting the Mixing Behavior of Aqueous Solutions Using a Machine Learning Framework," *ACS Applied Materials & Interfaces* 13 (2021): 11449–11460, https://pubs.acs.org/doi/10.1021/acsami.0c21036.

37. T. B. Martin and D. J. Audus, "Emerging Trends in Machine Learning: A Polymer Perspective," *ACS Polymers Au* 3 (2023): 239–258, https://doi.org/10.1021/acspolymersau.2c00053.

38. T. Meng, X. Jing, Z. Yan, and W. Pedrycz, "A Survey on Machine Learning for Data Fusion," *Information Fusion* 57 (2020): 115–129, https://doi.org/10.1016/j.inffus.2019.12.001.

39. A. Jha, A. Chandrasekaran, C. Kim, and R. Ramprasad, "Impact of dataset uncertainties on machine learning model predictions: the example of polymer glass transition temperatures," *Modelling and Simulation in Materials Science and Engineering* 27 (2019), https://doi.org/10.1088/1361-651X/aaf8ca.

40. S. Otsuka, I. Kuwajima, J. Hosoya, Y. Xu, and M. Yamazaki, Polymer Database for Polymeric Materials Design, International Conference on Emerging Intelligent Data and Web Technologies (EIDWT) (2011): 22–29, https://doi.org/10.1109/EIDWT.2011.13.

41. R. Ma and T. Luo, "PI1M: A Benchmark Database for Polymer Informatics," *Journal of Chemical Information and Modeling* 60 (2020): 4684–4690, https://doi.org/10.1021/acs.jcim.0c00726.

42. T. D. Huan, A. Mannodi-Kanakkithodi, C. Kim, V. Sharma, G. Pilania, and R. Ramprasad, "A Polymer Dataset for Accelerated Property Prediction and Design," in *Scientific Data* 3 (2016): 160012, https://doi.org/10.1038/sdata.2016.12.

43. S. Kim, J. Chen, T. Cheng, et al., "PubChem 2025 update," *Nucleic Acids Research* 53 (2024): D1516–D1525, https://doi.org/10.1093/nar/gkae1059.

44. J.-F. Wang, Y.-B. Sun, Q.-T. Chen, et al., "OpenPoly: A Polymer Database Empowering Benchmarking and Multi-property Predictions," *Chinese Journal of Polymer Science* 43 (2025): 1749–1760, https://doi.org/10.1007/s10118-025-3402-y.

45. Z. Sodaei, S. Ekrami, and S. M. Hashemianzadeh, "Machine Learning Analysis of Molecular Dynamics Properties Influencing Drug Solubility," *Scientific Reports* 15 (2025): 26955, https://doi.org/10.1038/s41598-025-11392-1.

46. "Pearson's Correlation Coefficient," In *Encyclopedia of Public Health*, (W. Kirch, Ed.) (Springer Netherlands, 2008): 1090–1091, https://doi.org/10.1007/978-1-4020-5614-7_2569.

47. S. Wold, K. Esbensen, and P. Geladi, "Principal Component Analysis," *Chemometrics and Intelligent Laboratory Systems* 2 (1987): 37–52, https://doi.org/10.1016/0169-7439(87)80084-9.

48. M. J. Uddin and J. Fan, "Interpretable Machine Learning Framework to Predict the Glass Transition Temperature of Polymers," *Polymers* 16 (2024): 1049, https://doi.org/10.3390/polym16081049.

49. Y. Xu, S. Zomer, and R. G. Brereton, "Support Vector Machines: A Recent Method for Classification in Chemometrics," *Critical Reviews in Analytical Chemistry* 36 (2006): 177–188, https://doi.org/10.1080/10408340600969486.

50. J. Zhao, Z. Li, Q. Gao, et al., "A Review of Statistical Methods for Dietary Pattern Analysis," *Nutrition Journal* 20 (2021): 37, https://doi.org/10.1186/s12937-021-00692-7.

51. V. L. Deringer, A. P. Bartók, N. Bernstein, D. M. Wilkins, M. Ceriotti, and G. Csányi, "Gaussian Process Regression for Materials and Molecules," *Chemical Reviews* 121 (2021): 10073–10141, https://doi.org/10.1021/acs.chemrev.1c00022.

52. N. Artemenko, I. Baskin, V. Palyulin, and N. Zefirov, "Prediction of Physical Properties of Organic Compounds Using Artificial Neural

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Networks Within the Substructure Approach," *Doklady Chemistry* 381 (2001): 317–320, https://doi.org/10.1023/A:1012976623974.

53. M. Meunier and S. Robertson, "Materials Studio 20th Anniversary," *Molecular Simulation* 47 (2021): 537–539, https://doi.org/10.1080/08927022.2021.1892093.

54. R. L. C. Akkermans, N. A. Spenley, and S. H. Robertson, "COMPASS III: Automated Fitting Workflows and Extension to Ionic Liquids," *Molecular Simulation* 47 (2021): 540–551, https://doi.org/10.1080/08927022.2020.1808215.

55. F. Furche, R. Ahlrichs, C. Hättig, W. Klopper, M. Sierka, and F. T. Weigend, "Turbomole," *WIREs Computational Molecular Science* 4 (2014): 91–100, https://doi.org/10.1002/wcms.1162.

56. G. V. Fabian Pedregosa, A. Gramfort, V. Michel, et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research* 12 (2011): 2825–2830.

57. C. R. Harris, K. J. Millman, S. J. van der Walt et al., "Array programming with NumPy," *Nature* 585 (2020): 357–362, https://doi.org/10.1038/s41586-020-2649-2.

58. W. McKinney, "Pandas: A Foundational Python Library for Data Analysis and Statistics," *Python High Performance Science Computer* 14 (2011): 1–9, https://docslib.org/doc/4231522/a-foundational-python-library-for-data-analysis-and-statistics.

59. P. Barrett, J. Hunter, J. T. Miller, J.-C. Hsu, and P. Greenfield, "matplotlib — A Portable Python Plotting Package," in *Astronomical Data Analysis Software and Systems XIV*, 2005.

60. P. Virtanen, R. Gommers, T. E. Oliphant, et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods* 17 (2020): 261–272, https://doi.org/10.1038/s41592-019-0686-2.

61. D. S. Wigh, J. M. Goodman, and A. A. Lapkin, "A Review of Molecular Representation in the Age of Machine Learning," *WIREs Computational Molecular Science* 12 (2022): 1603, https://doi.org/10.1002/wcms.1603.

62. T. Zhang, H. Li, H. Xi, R. V. Stanton, and S. H. Rotstein, "HELM: A Hierarchical Notation Language for Complex Biomolecule Structure Representation," *Journal of Chemical Information and Modeling* 52 (2012): 2796–2806, https://pubs.acs.org/doi/10.1021/ci3001925.

63. A. Drefahl, "CurlySMILES: A Chemical Language to Customize and Annotate Encodings of Molecular and Nanodevice Structures," *Journal of Cheminformatics* 3 (2011): 1, https://doi.org/10.1186/1758-2946-3-1.

64. T.-S. Lin, C. W. Coley, H. Mochigase, et al., "BigSMILES: A Structurally-Based Line Notation for Describing Macromolecules," *ACS Central Science* 5 (2019): 1523–1531, https://pubs.acs.org/doi/10.1021/acscentsci.9b00476.

65. M. Aldeghi and C. W. Coley, "A Graph Representation of Molecular Ensembles for Polymer Property Prediction," *Chemical Science* 13 (2022): 10486–10498, https://doi.org/10.1039/D2SC02839E.

66. D. N. Theodorou and U. W. Suter, "Detailed Molecular Structure of a Vinyl Polymer Glass," *Macromolecules* 18 (1985): 1467–1478, https://doi.org/10.1021/ma00149a018.

67. M. Kouza and U. H. Hansmann, "Velocity Scaling for Optimizing Replica Exchange Molecular Dynamics," *The Journal of Chemical Physics* 134 (2011): 044124, https://doi.org/10.1063/1.3533236.

68. H. J. C. Berendsen, J. P. M. Postma, W. F. van Gunsteren, A. DiNola, and J. R. Haak, "Molecular Dynamics with Coupling to an External Bath," *The Journal of Chemical Physics* 81 (1984): 3684–3690, https://doi.org/10.1063/1.448118.

69. M. Parrinello and A. Rahman, "Crystal Structure and Pair Potentials: A Molecular-Dynamics Study," *Physical Review Letters* 45 (1980): 1196–1199, https://doi.org/10.1103/PhysRevLett.45.1196.

70. S. Nosé, "A Unified Formulation of the Constant Temperature Molecular Dynamics Methods," *The Journal of Chemical Physics* 81 (1984): 511–519, https://doi.org/10.1063/1.447334.

71. W. G. Hoover, "Canonical Dynamics: Equilibrium Phase-space Distributions," *Physical Review A* 31 (1985): 1695–1697, https://doi.org/10.1103/PhysRevA.31.1695.

72. C. W. Yap, "PaDEL-Descriptor: An Open Source Software to Calculate Molecular Descriptors and Fingerprints," *Journal of Computational Chemistry* 32 (2011): 1466–1474, https://doi.org/10.1002/jcc.21707.

73. N. Van Nhu, M. Singh, and K. Leonhard, "Quantum Mechanically Based Estimation of Perturbed-Chain Polar Statistical Associating Fluid Theory Parameters for Analyzing Their Physical Significance and Predicting Properties," *The Journal of Physical Chemistry B* 112 (2008): 5693–5701, https://doi.org/10.1021/jp7105742.

74. J. Zheng, X. Xu, and D. G. Truhlar, "Minimally Augmented Karlsruhe Basis Sets," *Theoretical Chemistry Accounts* 128 (2011): 295–305, https://doi.org/10.1007/s00214-010-0846-z.

75. S. Grimme, S. Ehrlich, and L. Goerigk, "Effect of the Damping Function in Dispersion Corrected Density Functional Theory," *Journal of Computational Chemistry* 32 (2011): 1456–1465, https://doi.org/10.1002/jcc.21759.

76. A. D. Buckingham, R. L. Disch, and D. A. Dunmur, "Quadrupole Moments of Some Simple Molecules," *Journal of the American Chemical Society* 90 (1968): 3104–3107, https://doi.org/10.1021/ja01014a023.

77. J. Bicerano, *Prediction of Polymer Properties*, 3rd ed. (Press, C, 2002), https://doi.org/10.1201/9780203910115.

78. S. B. Dworakowska, "Application of Connectivity Indices in Polymer Chemistry," in *Proceedings of the 16th International Electronic Conference on Synthetic Organic Chemistry* (2012), https://doi.org/10.3390/ecsoc-16-01129.

79. S. Sharma, P. Kumar, and R. Chandra, "Chapter 7 – Applications of BIOVIA Materials Studio, LAMMPS, and GROMACS in Various Fields of Science and Engineering," in *Molecular Dynamics Simulation of Nanocomposites Using BIOVIA Materials Studio, Lammps and Gromacs*, (Ed. S. Sharma), (Elsevier, 2019), 329–341, https://doi.org/10.1016/B978-0-12-816954-4.00007-3.

80. D. Berrar, "Cross-Validation," in *Encyclopedia of Bioinformatics and Computational Biology*, (Ed. S. Ranganathan, M. Gribskov, K. Nakai, and C. Schönbach), (Academic Press, 2019), 542–545, https://doi.org/10.1016/B978-0-12-809633-8.20349-X.

81. C. E. Rasmussen, "Gaussian Processes in Machine Learning," in *Advanced Lectures on Machine Learning*, O. Bousquet, U. von Luxburg, and G. Rätsch, Eds., Lecture Notes in Computer Science 3176 (2004): 63–71, https://doi.org/10.1007/978-3-540-28650-9_4.

82. K. G. Joback and R. C. Reid, "Estimation of Pure-Component Properties from Group-Contributions," *Chemical Engineering Communications* 57 (1987): 233–243, https://doi.org/10.1080/00986448708960487.

83. S. Abdi, K. Movagharnejad, and H. Ghasemitabar, "Estimation of the Enthalpy of Vaporization at Normal Boiling Temperature of Organic Compounds by a New Group Contribution Method," *Fluid Phase Equilibria* 473 (2018): 166–174, https://doi.org/10.1016/j.fluid.2018.06.006.

84. C. Lin, L. Liu, Y. Liu, and J. Leng, "The Compatibility of Polylactic Acid and Polybutylene Succinate Blends by Molecular and Mesoscopic Dynamics," *International Journal of Smart and Nano Materials* 11 (2020): 24–37, https://doi.org/10.1080/19475411.2020.1729274.

85. A. Rheineck and K. Lin, "Solubility Parameter Calculations Based on Group Contributions," *Journal of Paint Technology* 40 (1968): 611–616.

86. C. W. Bunn, "The Melting Points of Chain Polymers," *Journal of Polymer Science* 16 (1955): 323–343, https://doi.org/10.1002/pol.1955.120168222.

87. H. Ahmad and M. Yaseen, "Application of a Chemical Group Contribution Technique for Calculating Solubility Parameters of Polymers," *Polymer Engineering & Science* 19 (1979): 858–863, https://doi.org/10.1002/pen.760191208.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

88. S. Kunchapu and K. M. Jablonka, "PolyMetriX: An Ecosystem for Digital Polymer Chemistry," *npj Computational Materials* 11 (2025): 312, https://doi.org/10.1038/s41524-025-01823-y.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adts70375-sup-0001-SuppMat.pdf.

25130390, 2026, 3, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adts.202501865 by Thuringer Universitats- Und, Wiley Online Library on [22/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License